
MA 553: Mathematical Foundations of Financial Engineering

Course Code	MA 553
Course Name	Mathematical Foundations of Financial Engineering
L-T-P-C	2-1-0-3
Prerequisites	IC 110, IC 111, IC 210
Intended for	3rd and 4th Year UG/PG
Distribution	Elective
One-time approval	9th BoA (Oct 2015); 9th Senate (May 2016), in the M.Sc Applied Mathematics curriculum

Course Contents

- **Module I** [3 Lectures]
Convex Analysis, constraint qualifications for convex optimization. Numerical Solution of QPP.
- **Module II** [15 Lectures]
Riemann Integration, Measurable space and function, Lebesgue Integration, Wiener process and properties, Martingales and stopping times, Strong Markov property, stochastic integrals, Ito processes.
- **Module III** [12 Lectures]
Introduction to PDE, Diffusion equation and its numerical solution, Diffusion process, connecting stochastic and partial differential equations, Black-Scholes Equation, numerical solution of Black-Scholes Equation.

Text Books

1. Thomas Mikosch, **Elementary stochastic Calculus with Finance in View**, World Scientific, 1999.
2. M. Capinski and T. Zastawniak, **Mathematics for finance: an introduction to financial engineering**, Springer, 2010.
3. Tavella, Domingo, and Curt Randall, **Pricing financial instruments: The finite difference method**, Vol. 13. John Wiley & Sons, 2000.

References

1. M. Baxter and A. Rennie, **Financial Calculus**, Cambridge University Press, 1996.
2. Olvi L. Mangasarian, **Nonlinear Programming**, Society for Industrial and Applied Mathematics (SIAM), 1994.
3. Walter Rudin, **Principles of Mathematical Analysis**, 3rd Edition, McGraw Hill, 1976.
4. S. N. Neftci, **Principles of financial engineering**, Academic Press/ Elsevier, 2009.